Contents

Q1 First Order Upwind

```
clc
clear
dtdx=0.8; % dt/dx
inter=[-1,1]; % interval
points=80; % number of points
dom=linspace(inter(1),inter(2),points); % interval broken into points
% first order upwind
V=-sin(pi*dom); % intial condition
Vi=V;
tim=30;
n=ceil(tim/(0.8*(dom(2)-dom(1)))); % number of times to be updated
for i=1:n
    % A be forward difference and B to be backward
    A(1:points-1)=V(2:points)-V(1:points-1);
    A(points)=V(1)-V(points); % periodic boundary
    B(2:points)=V(2:points)-V(1:points-1);
    B(1)=V(1)-V(points);
    sigmaA(1:points)=V.^0;sigmaB(1:points)=V.^0;
    signA=zeros(points,1)';signB=zeros(points,1)';
    for i2=1:points
        if sigmaA(i2)<=0
            signA(i2)=1;
        end
        if sigmaB(i2)>=0
            signB(i2)=1;
        end
    end
    V=V-dtdx*(A.*signA+B.*signB);
end

%plot(dom,V,'g')
%hold on
%plot(dom,-sin(pi*(dom-tim)),'r')
```

Q1 Second Order

```
clc
clear
dtdx=0.8; % dt/dx
inter=[-1,1]; % interval
points=80; % number of points
dom=linspace(inter(1),inter(2),points); % interval broken into points
% second order central form
V=-sin(pi*dom); % intial condition
Vi=V;
tim=30;
n=ceil(tim/(0.8*(dom(2)-dom(1)))); % number of times to be updated
for i=1:n
    % A be forward difference and B to be backward
    A(1:points-1)=V(2:points)-V(1:points-1);
    A(points)=V(1)-V(points); % periodic boundary
    B(2:points)=V(2:points)-V(1:points-1);
```

```
    B(1)=V(1)-V(points);
    sigmaA(1:points)=V.^0;sigmaB(1:points)=V.^0;
    fracA=dtdx*sigmaA;fracB=dtdx*sigmaB;
    V=V-dtdx*(A.*((1-fracA)/2)+B.*((1+fracB)/2));
end

%plot(dom,V,'g')
%hold on
%plot(dom,-sin(pi*(dom-tim)),'r')
```

Traffic Green

```
clc
clear
dtdx=0.8; % dt/dx
inter=[-30,20]; % interval
points=100; % number of points
dom=linspace(inter(1),inter(2),points); % interval broken into points
% first order upwind
in_zero=ceil(30*(points-1)/50); %index where first zero
V(1:in_zero)=ones(in_zero,1)';V(in_zero+1)=0.5;V(in_zero+2:points)=zeros(points-in_zero-1,1)';% intial condition
Vi=V;
tim=18;
n=ceil(tim/(0.8*(dom(2)-dom(1)))); % number of times to be updated
for i=1:n
    % A be forward difference and B to be backward
    FA(1:points-1)=V(2:points).*(1-V(2:points)) - V(1:points-1).*(1-V(1:points-1));
    FA(points)=0;
    FB(2:points)=V(2:points).*(1-V(2:points)) - V(1:points-1).*(1-V(1:points-1));
    FB(1)=0;

    A(1:points-1)=V(2:points) - V(1:points-1);
    A(points)=0;
    B(2:points)=V(2:points) - V(1:points-1);
    B(1)=0;

    % Since we cant divide by zero so Sign(x/y) is replaced by Sign(x*y)
    sigmaA(1:points)=FA.*A;sigmaB(1:points)=FB.*B;
    signA=zeros(points,1)';signB=zeros(points,1)';
    for i2=1:points
        if sigmaA(i2)<=0
            signA(i2)=1;
        end
        if sigmaB(i2)>=0
            signB(i2)=1;
        end
    end
    V=V-dtdx*(FA.*signA+FB.*signB);
end
%plot(dom,Vi,'r')
%hold on
%plot(dom,V,'g')


% Second order central Form
V=Vi;
for i=1:n
    % A be forward difference and B to be backward
    % A be forward difference and B to be backward
    FA(1:points-1)=V(2:points).*(1-V(2:points)) - V(1:points-1).*(1-V(1:points-1));
    FA(points)=0;
    FB(2:points)=V(2:points).*(1-V(2:points)) - V(1:points-1).*(1-V(1:points-1));
    FB(1)=0;

    aA(1:points-1)=1-(V(2:points) + V(1:points-1));aA(points)=1-2*V(points);
```

```
    aB(2:points)=1-(V(2:points) + V(1:points-1));aB(1)=1-2*V(1);

    fracA=dtdx*aA;fracB=dtdx*aB;

    V=V-dtdx*(FA.*((1-fracA)/2)+FB.*((1+fracB)/2));
end

%plot(dom,Vi,'r')
%hold on
%plot(dom,V,'g')
%plot(dom,V)
```

Traffic Red

```
clc
clear
dtdx=0.8; % dt/dx
inter=[-30,20]; % interval
points=100; % number of points
dom=linspace(inter(1),inter(2),points); % interval broken into points
% first order upwind
in_zero=ceil(30*(points-1)/50); %index where first zero
V(1:in_zero)=0.25*ones(in_zero,1)';V(in_zero+1)=0.25;V(in_zero+2:points)=ones(points-in_zero-1,1)';% intial condition
Vi=V;
tim=36;
n=ceil(tim/(0.8*(dom(2)-dom(1)))); % number of times to be updated
for i=1:n
    % A be forward difference and B to be backward
    FA(1:points-1)=V(2:points).*(1-V(2:points)) - V(1:points-1).*(1-V(1:points-1));
    FA(points)=0;
    FB(2:points)=V(2:points).*(1-V(2:points)) - V(1:points-1).*(1-V(1:points-1));
    FB(1)=0;

    A(1:points-1)=V(2:points) - V(1:points-1);
    A(points)=0;
    B(2:points)=V(2:points) - V(1:points-1);
    B(1)=0;

    % Since we cant divide by zero so Sign(x/y) is replaced by Sign(x*y)
    sigmaA(1:points)=FA.*A;sigmaB(1:points)=FB.*B;
    signA=zeros(points,1)';signB=zeros(points,1)';
    for i2=1:points
        if sigmaA(i2)<=0
            signA(i2)=1;
        end
        if sigmaB(i2)>=0
            signB(i2)=1;
        end
    end
    V=V-dtdx*(FA.*signA+FB.*signB);
end
%plot(dom,Vi,'r')
%hold on
%plot(dom,V,'g')
%plot(dom,V)

% Second order central Form
V=Vi;
for i=1:n
    % A be forward difference and B to be backward
    % A be forward difference and B to be backward
    FA(1:points-1)=V(2:points).*(1-V(2:points)) - V(1:points-1).*(1-V(1:points-1));
    FA(points)=0;
    FB(2:points)=V(2:points).*(1-V(2:points)) - V(1:points-1).*(1-V(1:points-1));
    FB(1)=0;
```

```
    aA(1:points-1)=1-(V(2:points) + V(1:points-1));aA(points)=1-2*V(points);
    aB(2:points)=1-(V(2:points) + V(1:points-1));aB(1)=1-2*V(1);

    fracA=dtdx*aA;fracB=dtdx*aB;

    V=V-dtdx*(FA.*((1-fracA)/2)+FB.*((1+fracB)/2));
end

%plot(dom,Vi,'r')
%hold on
%plot(dom,V,'g')
%plot(dom,V)
```

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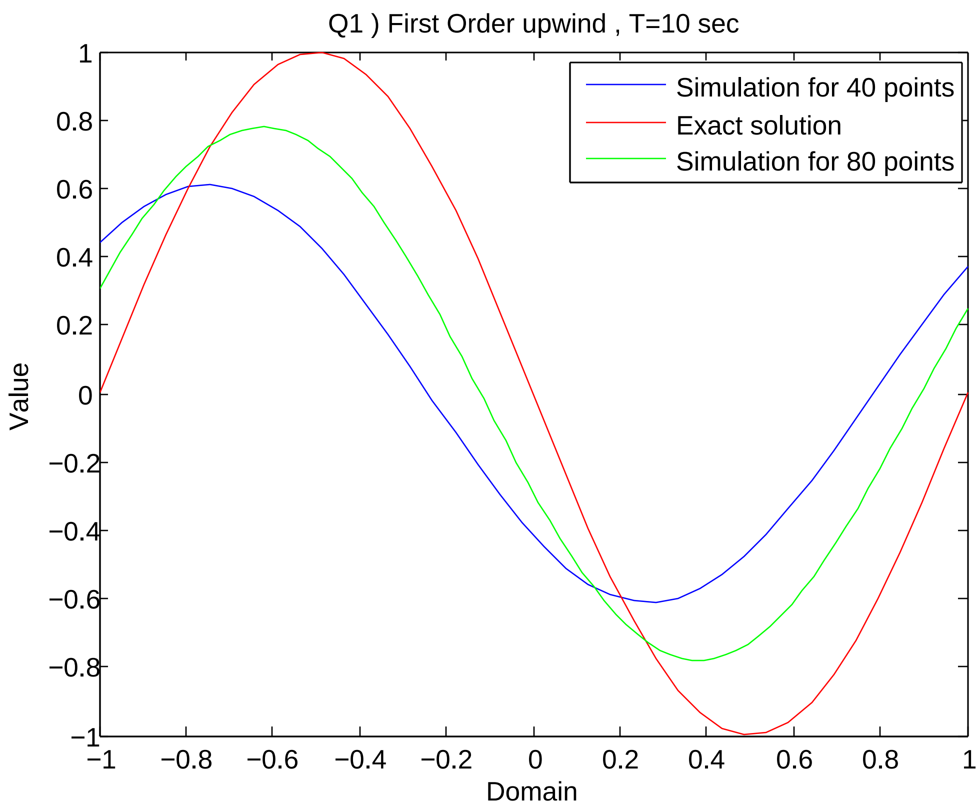
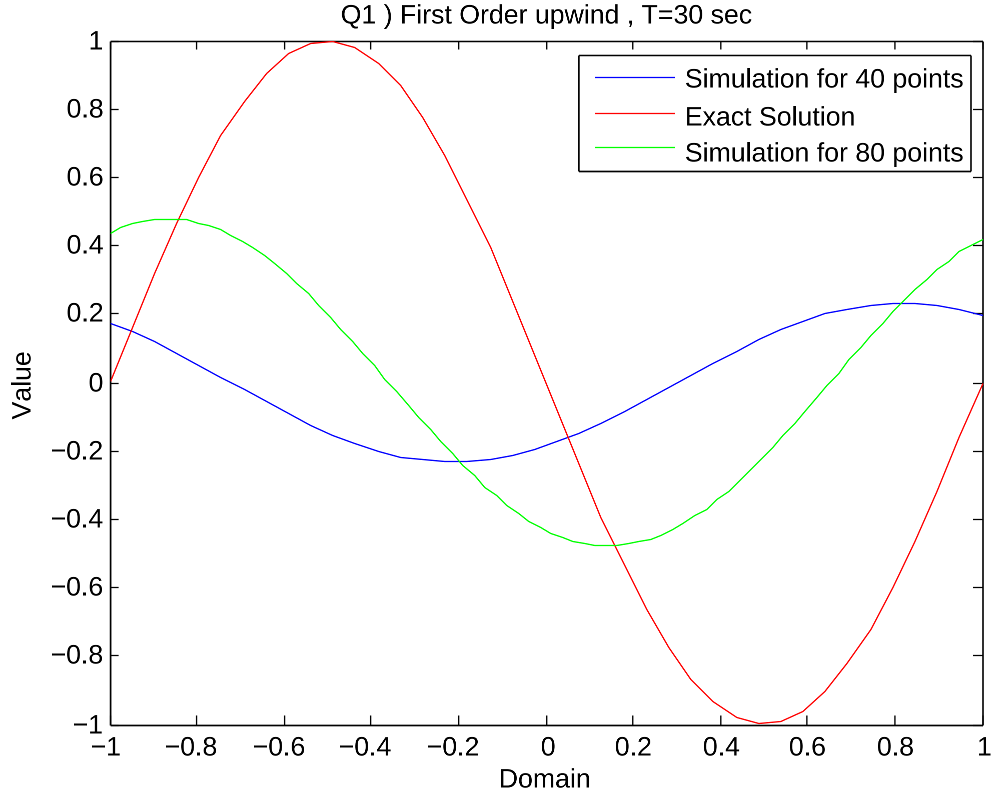
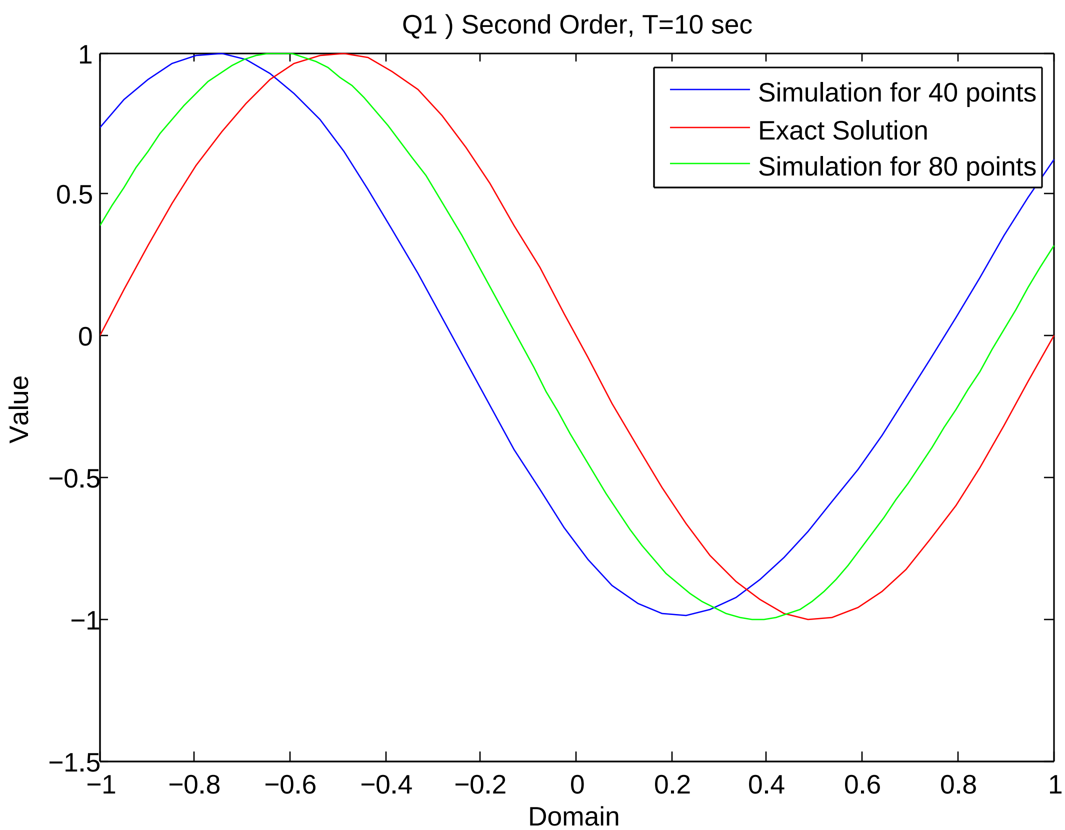
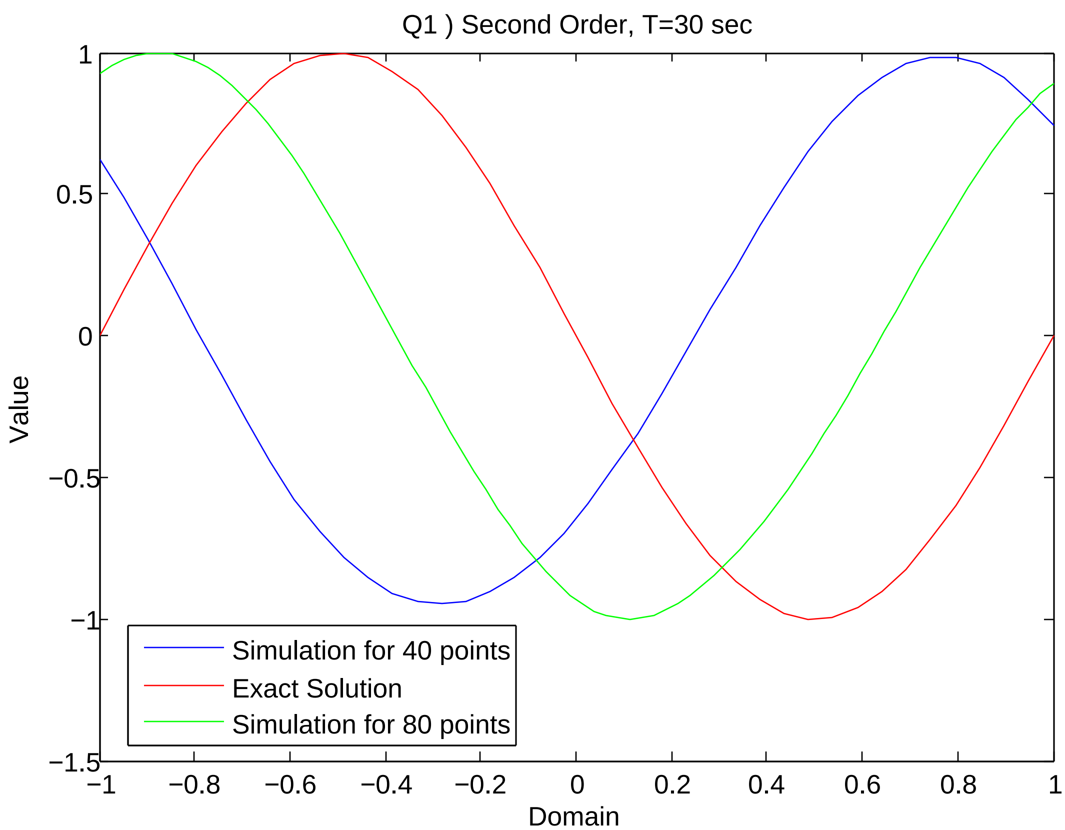

Q1) First Order upwind , T=10 sec
Simulation for 40 points
Exact solution
Simulation for 80 points
Value
Domain
1
0.8
0.6
0.4
0.2
0
−0.2
−0.4
−0.6
−0.8
−1
−1
−0.8
−0.6
−0.4
−0.2
0
0.2
0.4
0.6
0.8
1

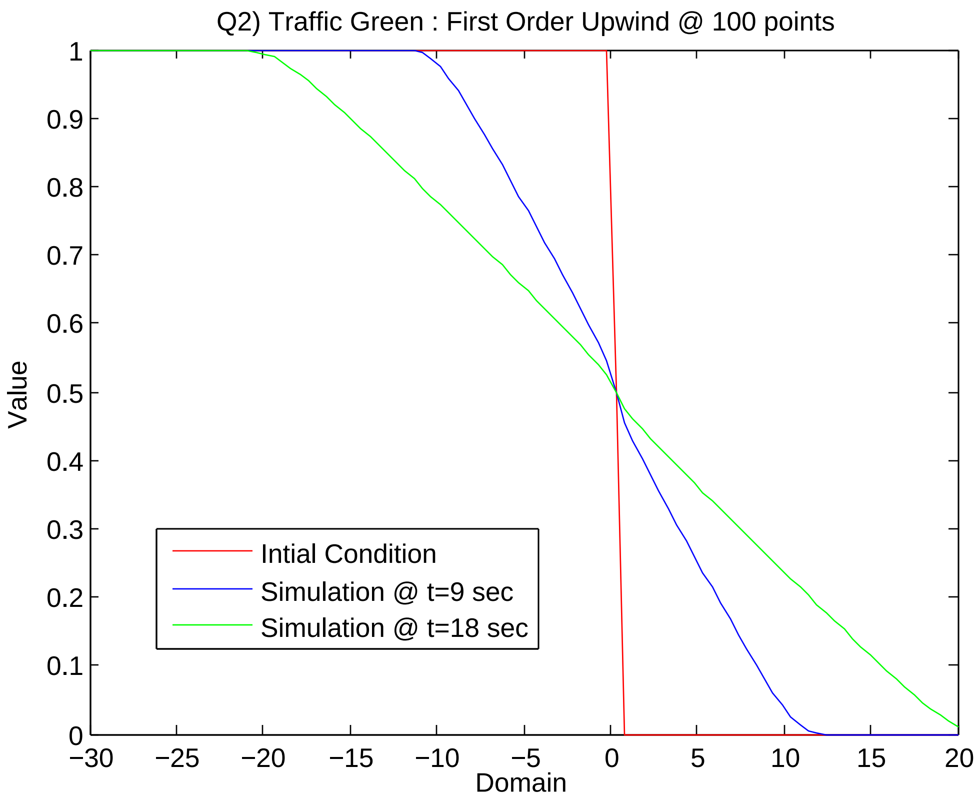
Q1) First Order upwind , T=30 sec

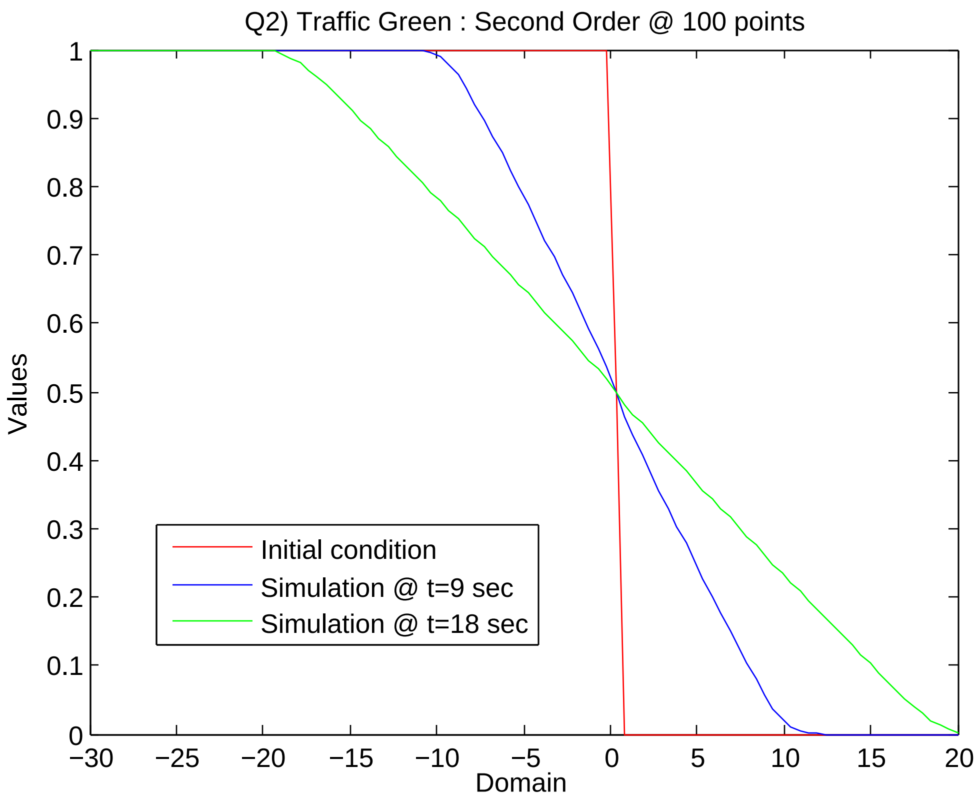


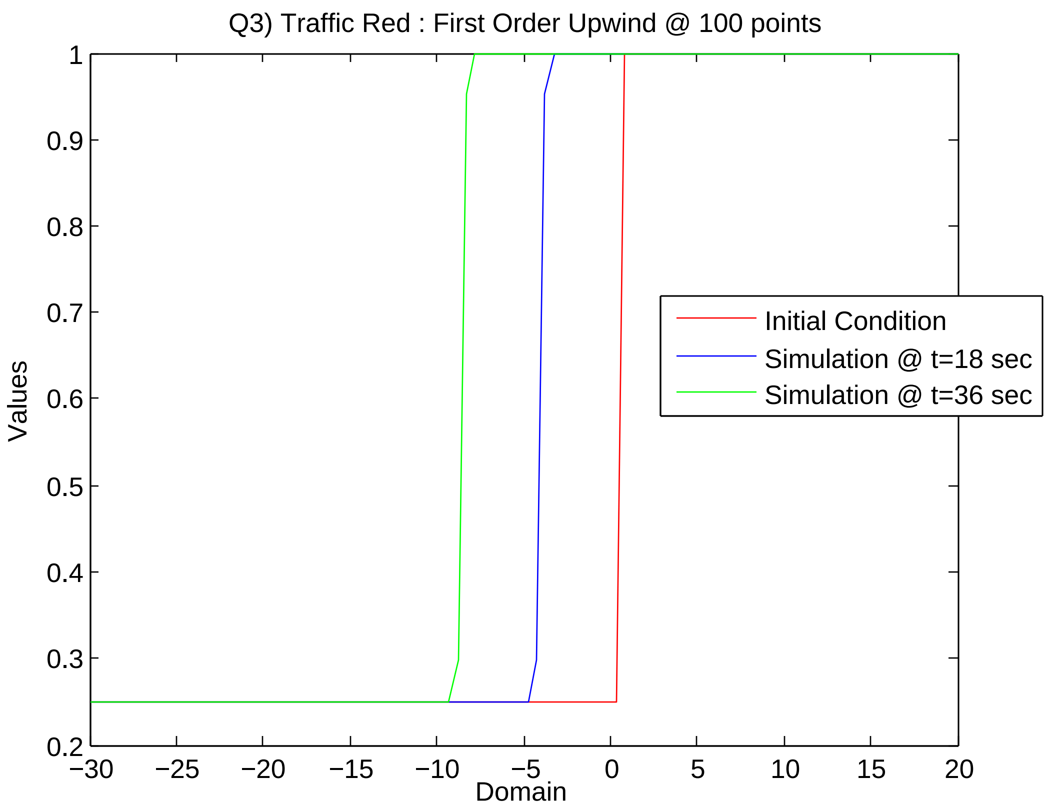


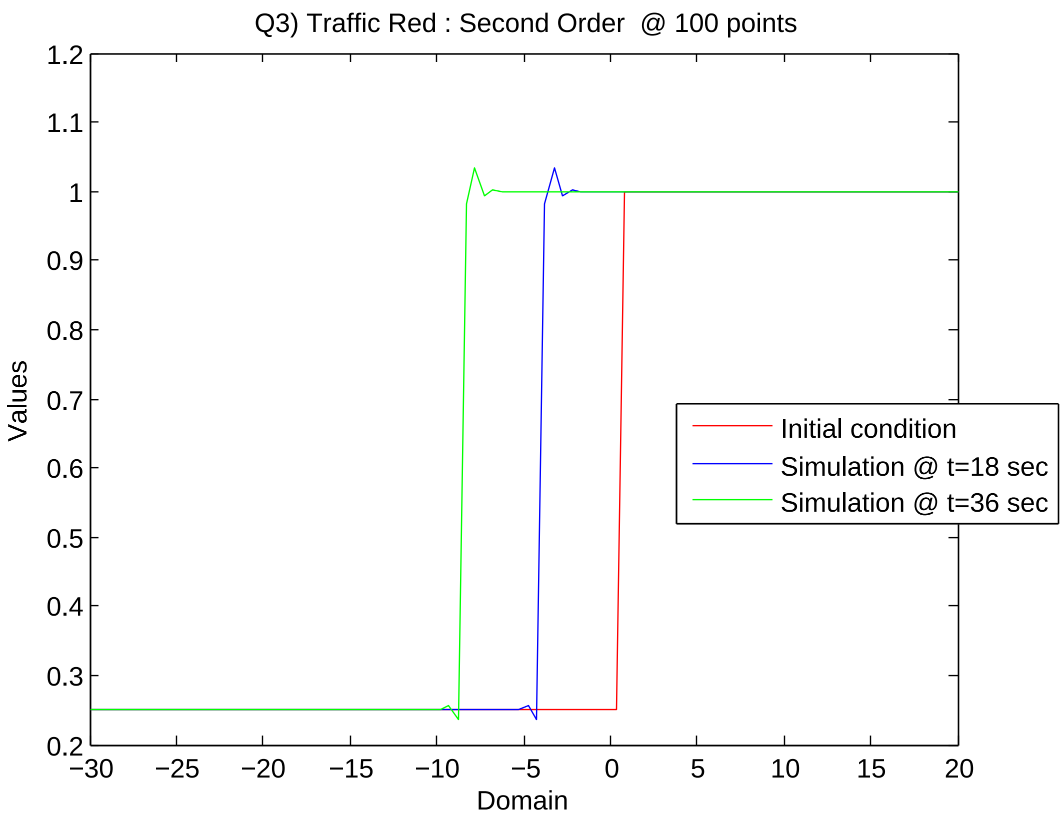
Q1) Second Order, T=10 sec
Simulation for 40 points
Exact Solution
Simulation for 80 points
Value
Domain
1
0.5
0
−0.5
−1
−1.5
−1
−0.8
−0.6
−0.4
−0.2
0
0.2
0.4
0.6
0.8
1


Q1) Second Order, T=30 sec
Value
Domain
1
0.5
0
−0.5
−1
−1.5
−1
−0.8
−0.6
−0.4
−0.2
0
0.2
0.4
0.6
0.8
1
Simulation for 40 points
Exact Solution
Simulation for 80 points


Q2) Traffic Green : First Order Upwind @ 100 points
Value
Domain
1
0.9
0.8
0.7
0.6
0.5
0.4
0.3
0.2
0.1
0
−30
−25
−20
−15
−10
−5
0
5
10
15
20
Intial Condition
Simulation @ t=9 sec
Simulation @ t=18 sec


Q2) Traffic Green : Second Order @ 100 points
Values
Domain
1
0.9
0.8
0.7
0.6
0.5
0.4
0.3
0.2
0.1
0
−30
−25
−20
−15
−10
−5
0
5
10
15
20
Initial condition
Simulation @ t=9 sec
Simulation @ t=18 sec


Q3) Traffic Red : First Order Upwind @ 100 points
Values
Domain
Initial Condition
Simulation @ t=18 sec
Simulation @ t=36 sec
1
0.9
0.8
0.7
0.6
0.5
0.4
0.3
0.2
−30
−25
−20
−15
−10
−5
0
5
10
15
20


Q3) Traffic Red : Second Order @ 100 points
Values
Domain
1.2
1.1
1
0.9
0.8
0.7
0.6
0.5
0.4
0.3
0.2
−30
−25
−20
−15
−10
−5
0
5
10
15
20
Initial condition
Simulation @ t=18 sec
Simulation @ t=36 sec